Eco-friendly hydrogels: Optimizing *Ailanthus altissima* Biochar Content for Copper Adsorption

Introduction

1. Copper contamination is dangerous as increased exposure can lead to cancer (Abdelmonem et al., 2025).
2. Copper is **not biodegradable** and can affect plants processes like photosynthesis.
3. Wastewater treatment like reverse osmosis have limited productivity, **high cost** and create hazardous byproducts like sludge (Yi et al., 2025).
4. Adsorption is low cost, biodegradable, and from abundant polymers making a good alternative (Yi et al., 2025).

Why biochar?


(Skillings & Sons, 2015),


(Hassani, 2026)


(Gorny, 2025)

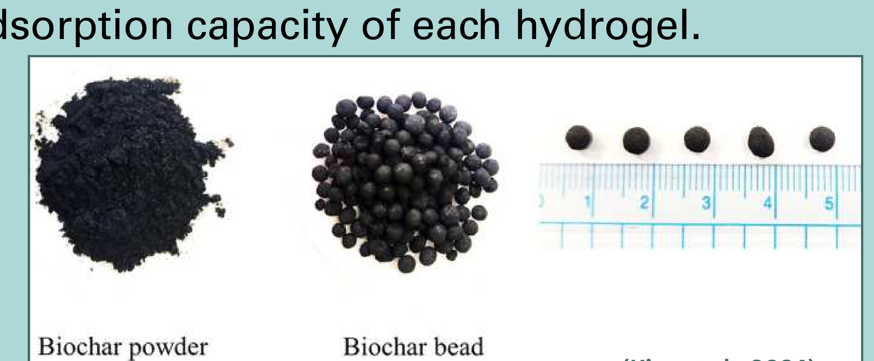
- Biochar is a **carbon** rich substance made from organic matter.
- Biochar adds **functional groups** and **porous structure** (Zhang et al., 2024).
- Prior research shows *Acacia dealbata* is a good biochar derivative due to **high lignin** content of 35% and its ability to grow quickly (Macena et al., 2026).
- *Ailanthus altissima* chokes out natives and invites spotted lantern flies (Soler & Izquierdo, 2024).
- Biochar from Tree-of-Heaven has unique **high** cellulose content (45%) **high** lignin (24%) and low ash (0.9%) which suggest **more** functional groups for copper bonding along with allelopathic chemicals (Terzopoulou et al., 2023).

Purpose

To study the effect of adding Tree-of-Heaven biochar in sodium alginate hydrogels (all ecofriendly materials), a popular adsorption technique, for copper removal in water.

IV: Sodium alginate to biochar ratio in hydrogel

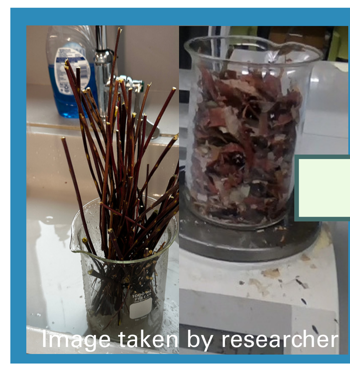
DV: Adsorption capacity of each hydrogel.



(Kim et al., 2024)

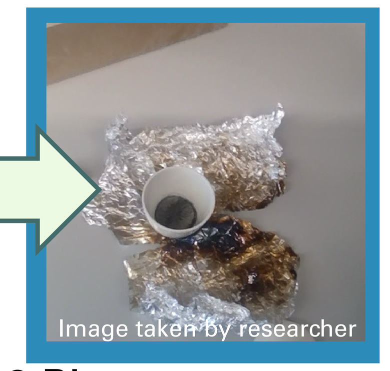
It was hypothesized that the a ratio of 4:1 sodium alginate to biochar would have the best **adsorption** as it creates a more porous structure due to **polymer** makeup and **allelopathic** chemicals, and more functional groups to allow for ionic bonding with positively charged heavy metals without crowding pores **excessively**.

Methodology



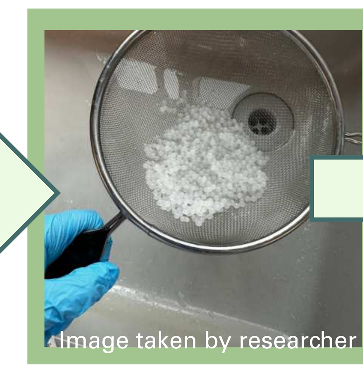
1.Tree-of-Heaven was collected and washed to remove impurities.

2. Biomass blended and sieve with 75 mesh.

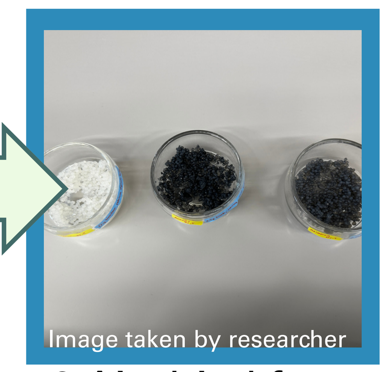


3.Biomass was put through pyrolysis for 2h, 500 C to create biochar (B).

4.Sonicated (30m) and added drop-wise in 0.2M CaCl2 150 mL.



5.Left in CaCl2 for 48 h then washed with DI water.

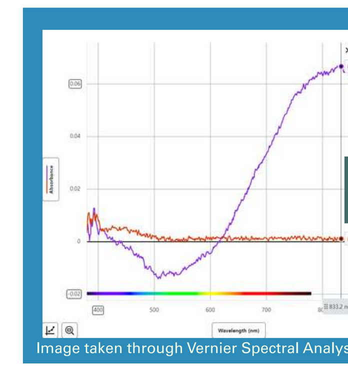


6.Air dried for 48 h for copper adsorption.

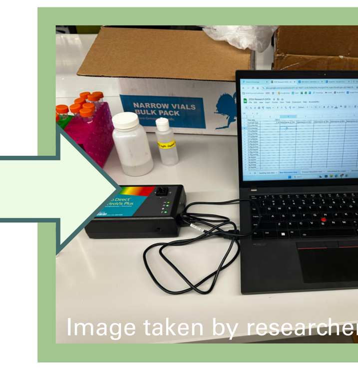
(Yi et al., 2025) (Du et al., 2023) (Wang et al., 2023) (Ayouch et al., 2020) (Chen et al., 2023)

Processing

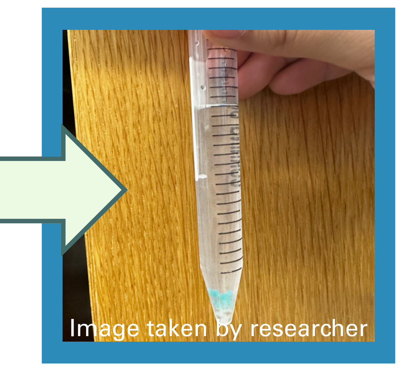
7. Swelling tests were assessed seeing volume gained over 48 hours.



8.UV-VIS spectroscopy was used for a calibration curve and Beer's Law (833.2nm).



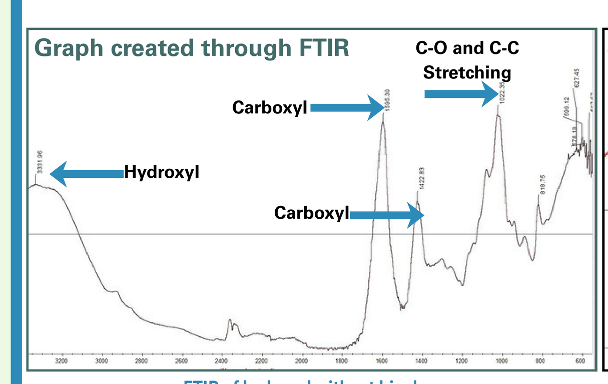
9.Adsorption was measured over 90 minutes, calculating adsorption capacity at intervals.

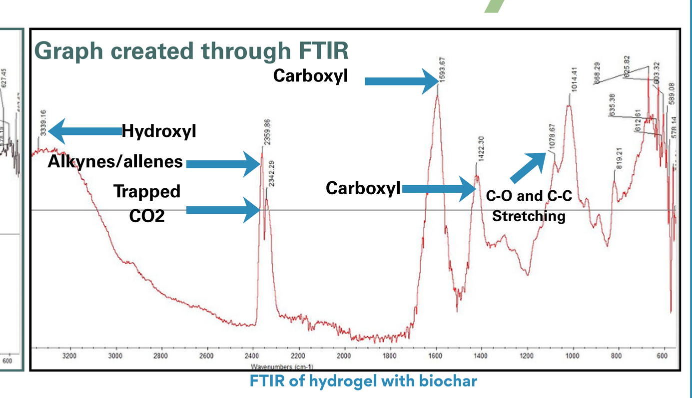


10.Hydrogels were dried and processed for FTIR and SEM analysis.

(Yi et al., 2025) Chen et al., 2023)

Characterization: FTIR Analysis





The FTIR analysis was used to determine if functional groups increased with biochar content.

FTIR with biochar has sharper and taller peaks show increased presence of functional groups (Yi et al., 2025).

Trapped CO2 is indicative of **porous** structure and alkynes and allenes indicate carbonization.

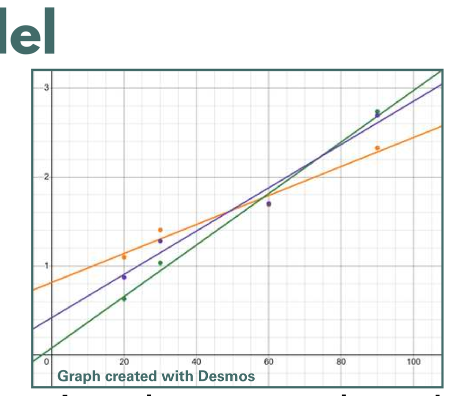
Characterization: Model and SEM

Pseudo first order model

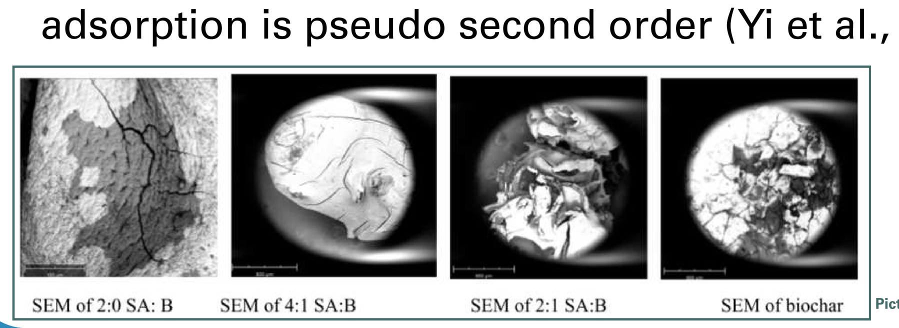
Pseudo first order model R squared of each respective hydrogel

Table created by researcher

Type of Hydrogel	R squared
2:0 SA:B	0.9688
4:1 SA:B	0.9696
2:1 SA:B	0.9904



It was assessed to see if the functional groups played a role in adsorption. Pseudo first order represents physical diffusion and Pseudo second order represents chemical bonding. A higher r^2 of time (t) versus t divided by adsorption is pseudo second order (Yi et al., 2025)

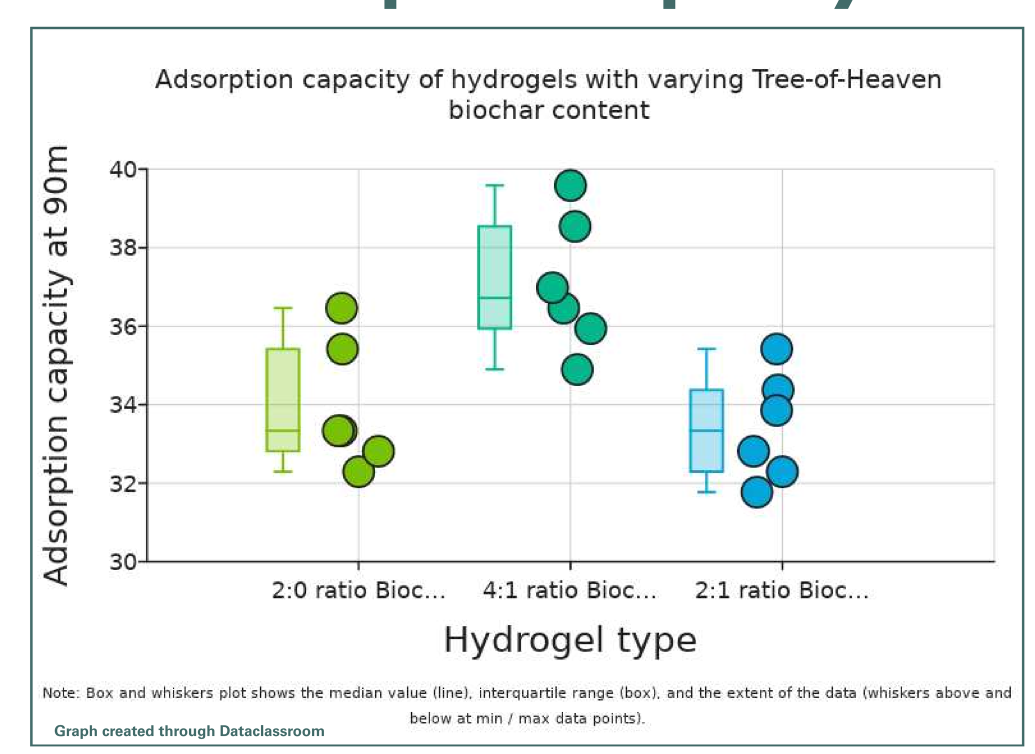


Pictures taken by research and SEM imaging

Shows increase in porosity with biochar.

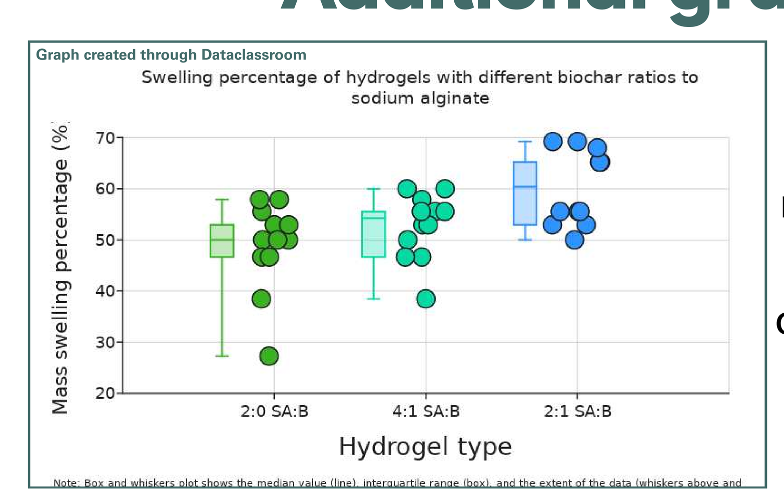
Results

Adsorption capacity

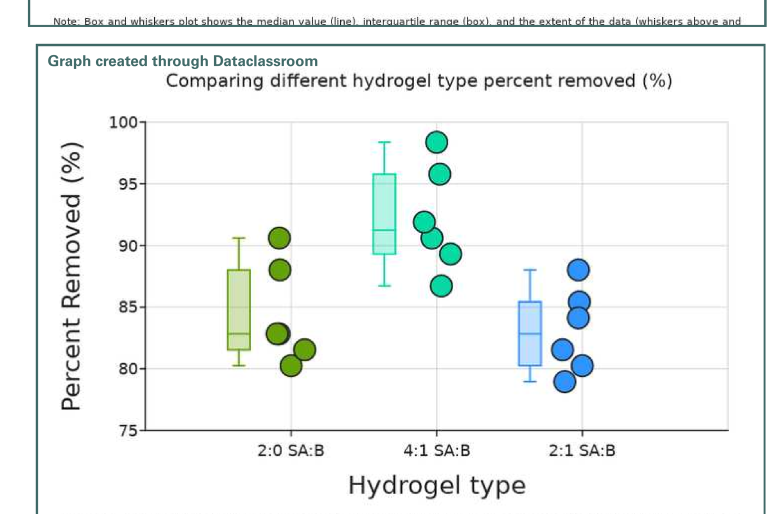


This graph shows adsorption capacity of each hydrogel Adsorption capacity is the amount of contaminant (mg) trapped by a dry hydrogel in grams (g).

Additional graphs



Mass swelling ratio shows material strength and ability to open to polymers for functional groups.



Percent removed shows the amount of copper removed from water by each respective hydrogel.

Conclusion

Statistical Analysis: Adsorption Capacity

Degrees of Freedom (df)	H-statistic	P-value
2	9.16	0.01
Pair	**P-value**	**Interpretation of P**
2:0 SA:B - 4:1 SA:B	0.07	Means: some confidence that the groups are different
2:0 SA:B - 2:1 SA:B	1	Means: that the groups are NOT different
4:1 SA:B - 2:1 SA:B	0.01	Means: very strong confidence that the groups are different

This p value suggests **very strong** confidence that the groups are difference.

Created through Google Sheets

Statistical Analysis: Swelling ratio

Degrees of Freedom (df)	H-statistic	P-value
2	9.01	**0.01**
Pair	**P-value**	**Interpretation of P**
2:0 SA:B - 4:1 SA:B	0.06	Means: some confidence that the groups are different
2:0 SA:B - 2:1 SA:B	1	Means: that the groups are NOT different
4:1 SA:B - 2:1 SA:B	0.01	Means: very strong confidence that the groups are different

This p value suggests **very strong** confidence that the groups are difference.

Created through Google Sheets

Statistical Analysis: Percent removed (%)

Degrees of Freedom (df)	H-statistic	P-value
2	9.63	0.01
Pair	**P-value**	**Interpretation of P**
2:0 ratio Biochar - 4:1	0.85	Means: no evidence that the groups might be different
2:0 ratio Biochar - 2:1	0.01	Means: very strong confidence that the groups are different
4:1 ratio Biochar - 2:1	0.14	Means: some indication that the groups might be

This p value suggests **very strong** confidence that the groups are difference.

Created through Google Sheets

The hydrogel adsorbs copper through chemisorption, utilizing **functional** groups and porous surfaces to bind to copper ions. Adding Tree-of-Heaven biochar enhances adsorption capacity (seen in 2:1 and 4:1). While this is a promising **sustainable** method for copper removal, **more** trials and ratios must be assessed.

Limitations/Extension

Some **limitations** include the restriction of temperature and pH. Other studies such as Yi et al., 2025 optimized the conditions for absorbance which could also be used for future research. There are also smaller trial groups making them susceptible for outliers.

This research could help support **additional additives** to biochar to maximize heavy metal adsorption. Future research can assess **more ratios** and see what external factors affect copper adsorption. Additionally, possible work can be done on **adsorbing dyes** as dyes are frequently found in the environment and wastewater.

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